

Chapter 2: Signal Theory

Exercise 1:

The continuous-time signal $x(t)$ is generated according to:

$$x(t) = \sum_{k=1}^3 b_k \cos(k\omega_0 t)$$
$$\text{with } \omega_0 = \frac{\pi}{5} \text{ rad/s,} \quad \text{and} \quad b_k = \begin{cases} \frac{2}{k\pi} & \text{for } k = 1, 3 \\ 0 & \text{for } k = 2 \end{cases}$$

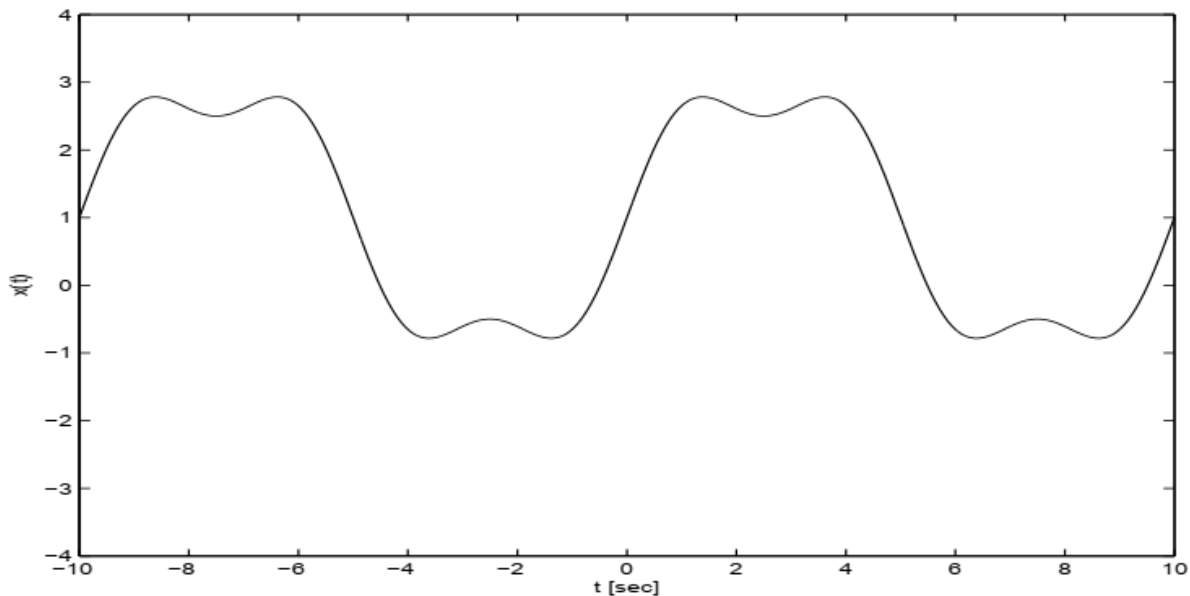
- Is $x(t)$ a periodic signal, and if so, what is its fundamental period?
- Is $x(t)$ an even or odd function? Motivate your answer.
- Determine the Fourier series of $x(t)$ in the complex exponential form. For this purpose, use the formula $\cos(\varphi) = \frac{1}{2}(e^{j\varphi} + e^{-j\varphi})$.
- Determine the power of the signal $x(t)$

Exercise 2:

In Figure 1, the infinite-time signal $x(t)$ is depicted over the interval $[-10, 10]$. This continuous-time signal $x(t)$ is periodic and is generated as

$$x(t) = a_0 + \sum_{k=1}^3 a_k \sin(k\omega_0 t)$$

with $\omega_0 = 2\pi/10$ rad/s, $a_0 = 1$, $a_1 = 2$, $a_2 = 0$ and $a_3 = 1$



- a. Is $x(t)$ an odd signal, an even signal or neither of the two? Motivate your answer.
- b. Determine the fundamental period of $x(t)$ and its fundamental frequency.
- c. Determine the Fourier series of $x(t)$ in the complex exponential form.
- d. Determine the power of the signal $x(t)$.